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Introduction

As part of my induction task at Urban Resource Intelligence (URI) - a consultancy that advises local authorities on improving the efficiency, sustainability, and resilience of urban resource systems, including housing, utilities, and environmental services, this report presents a mathematical investigation into urban residential analytics, utilizing linear algebra, calculus, probability, and statistics to examine resource demand patterns, system stability, uncertainty, and behavioural differences across residential groups. The investigation begins by modelling steady-state resource demand across four interconnected residential zones before moving to an analysis of how demand changes over time within a high-density area, an assessment of uncertainty in short-term forecasting, and a comparison of average daily consumption across different housing types. Each section sets out the technique applied, the calculations performed, and the practical significance of the findings for planning and policy decisions within local authorities.

Methodology

Question 1: Linear Algebra: Modelling Steady-State Resource Demand

Introduction

The problem is modelled as a system of linear equations solved using linear algebra, which is essential for managing interdependent multi-variable systems.

Approach

Gaussian elimination solves this system efficiently using elementary row operations, making it more effective than substitution for multiple-variable systems. (Strang, 2023)

Solution

The resource demand across zones one to four is defined by the following system of equations:

$$2x_1 - x_2 + x_3 = 120$$
$$-x_1 + 3x_2 - x_4 = 150$$
$$x_1 + x_2 + 2x_3 - x_4 = 180$$
$$-x_2 + x_3 + 2x_4 = 140$$

This can be expressed in the form Ax = b, where A is the coefficient matrix, x represents the variables, and b is the column vector with independent terms (Lay, et al., 2021).

$$\begin{bmatrix} 2 & -1 & 1 & 0 \\ -1 & 3 & 0 & -1 \\ 1 & 1 & 2 & -1 \\ 0 & -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 120 \\ 150 \\ 180 \\ 140 \end{bmatrix}$$

It can also be expressed as an augmented matrix as follows:

$$\left(\begin{array}{cccc|c} 2 & -1 & 1 & 0 & 120 \\ -1 & 3 & 0 & -1 & 150 \\ 1 & 1 & 2 & -1 & 180 \\ 0 & -1 & 1 & 2 & 140 \end{array}\right)$$

Let all rows be R_n, with n as the row number. R_1 is row one, up to R_4. Equations are solved using row operations until the augmented matrix's left side is in reduced row echelon form.:

$R_1 / 2 \rightarrow R_1$ (row 1 divided by 2 = row 1)

$$\begin{pmatrix} 1 & 0.5 & 0.5 & 0 & 60 \\ -1 & 3 & 0 & -1 & 150 \\ 1 & 1 & 2 & -1 & 180 \\ 0 & -1 & 1 & 2 & 140 \end{pmatrix}$$

$R_2 + R_1 \rightarrow R_2$ (row 2 plus row 1 = row 2)

$R_3 - R_1 \rightarrow R_3$ (row 3 minus row 1 = row 3)

$$\begin{pmatrix} 1 & 0.5 & 0.5 & 0 & 60 \\ 0 & 2.5 & 0.5 & -1 & 210 \\ 0 & 1.5 & 1.5 & -1 & 120 \\ 0 & -1 & 1 & 2 & 140 \end{pmatrix}$$

$R_2 / 2.5 \rightarrow R_2$ (row 2 divided by 2.5 = row 2)

$$\begin{pmatrix} 1 & 0.5 & 0.5 & 0 & 60 \\ 0 & 1 & 0.2 & -0.4 & 84 \\ 0 & 1.5 & 1.5 & -1 & 120 \\ 0 & -1 & 1 & 2 & 140 \end{pmatrix}$$

$R_1 + 0.5R_2 \rightarrow R_1$ (row 1 plus half of row 2 = row 1)

$R_3 - 1.5R_2 \rightarrow R_3$ (row 3 – row 2 multiplied by 1.5 = row 3)

$R_4 + R_2 \rightarrow R_4$ (row 4 plus row 2 = row 4)

$$\begin{pmatrix} 1 & 0 & 0.6 & -0.2 & 102 \\ 0 & 1 & 0.2 & -0.4 & 84 \\ 0 & 0 & 1.2 & -0.4 & -6 \\ 0 & 0 & 1.2 & 1.6 & 224 \end{pmatrix}$$

$R_3 / 1.2 \rightarrow R_3$ (row 3 divided by 1.2 = row 3)

$$\begin{pmatrix} 1 & 0 & 0.6 & -0.2 & 102 \\ 0 & 1 & 0.2 & -0.4 & 84 \\ 0 & 0 & 1 & -1/3 & -5 \\ 0 & 0 & 1.2 & 1.6 & 224 \end{pmatrix}$$

$R_1 - 0.6R_3 \rightarrow R_1$ (row 1 minus row 3 multiplied by 0.6 = row 1)

$R_2 - 0.2R_3 \rightarrow R_2$ (row 2 minus row 3 multiplied by 0.2 = row 2)

$R_4 - 1.2R_3 \rightarrow R_4$ (row 4 minus row 3 multiplied by 1.2 = row 4)

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 105 \\ 0 & 1 & 0 & -1/3 & 85 \\ 0 & 0 & 1 & -1/3 & -5 \\ 0 & 0 & 0 & 2 & 230 \end{pmatrix}$$

$R_4 / 2 \rightarrow R_4$ (half of row 4 = row 4)

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 105 \\ 0 & 1 & 0 & -1/3 & 85 \\ 0 & 0 & 1 & -1/3 & -5 \\ 0 & 0 & 0 & 1 & 115 \end{pmatrix}$$

$R_2 + 1/3R_4 \rightarrow R_2$ (row 2 plus one third of row 4 = row 2)

$R_3 + 1/3R_4 \rightarrow R_3$ (row 3 plus one third of row 4 = row 3)

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 105 \\ 0 & 1 & 0 & 0 & 370/3 \\ 0 & 0 & 1 & 0 & 100/3 \\ 0 & 0 & 0 & 1 & 115 \end{pmatrix}$$

∴

x_1 = 105,

x_2 = 370/3 or 123.33 (to two decimal places),

x_3 = 100/3 or 33.33 (to two decimal places),

x_4 = 115

Values are substituted into the original equations to verify that the left-hand side (LHS) equals the right-hand side (RHS) after evaluation.

$$2(105) – 370/3 + 100/3 = 120$$
$$-105 + 3(370/3) – 115 = 150$$
$$105 + 370/3 + 2(100//3) – 115 = 180$$
$$-370/3 + 100/3 + 2(115) = 140$$

The LHS and RHS are equal; thus, the equations are correctly solved.

Result

Residential Zone	Resource Demand (Units)
Zone 1 (x_1)	105.00
Zone 2 (x_2)	123.33
Zone 3 (x_3)	33.33
Zone 4 (x_4)	115.00

For a unique solution, A must be invertible, meaning its determinant must be non-zero. Computation uses cofactor expansion (Lay, et al., 2021) as shown below:

$$\boldsymbol{det(A) = 2M_{11} - (-1)M_{12} + 1M_{14} - 0M_{14}} \text{ where,}$$

$$M_{11} = \begin{vmatrix} 3 & 0 & -1 \\ 1 & 2 & -1 \\ -1 & 1 & 2 \end{vmatrix}, M_{12} = \begin{vmatrix} -1 & 0 & -1 \\ 1 & 2 & -1 \\ 0 & 1 & 2 \end{vmatrix}, M_{13} = \begin{vmatrix} -1 & 3 & -1 \\ 1 & 1 & -1 \\ 0 & -1 & 2 \end{vmatrix}, \text{ and } M_{14} = \begin{vmatrix} -1 & 3 & 0 \\ 1 & 1 & 2 \\ 0 & -1 & 1 \end{vmatrix}$$

So,

$$\mathbf{M_{11}} = 3\begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} - 0\begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix} + (-1)\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}$$

$$\Rightarrow 3[2(2) - (-1)(1)] - 0[(1)(2) - (-1)(-1)] -1[(1)(1) - (2)(-1)] = \mathbf{12}$$

$$\mathbf{M_{12}} = -1\begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} - 0\begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} + (-1)\begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix}$$

$$\Rightarrow -1[(2)(2) - (-1)(1)] - 0[(1)(2) - (-1)(0)] -1[(1)(1) - (2)(0)] = \mathbf{-6}$$

$$\mathbf{M_{13}} = -1\begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix} - 3\begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} + (-1)\begin{vmatrix} 1 & 1 \\ 0 & -1 \end{vmatrix}$$

$$\Rightarrow -1[(1)(2) - (-1)(-1)] -3[(1)(2) - (-1)(0)] -1[(1)(-1) - (1)(0)] = \mathbf{-6}$$

$$\mathbf{M_{14}} = -1\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} + 3\begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} + 0\begin{vmatrix} 1 & 1 \\ 0 & -1 \end{vmatrix}$$

$$\Rightarrow -1[(1)(1) - (2)(-1) + 3[(1)(1) - (2)(0)] + 0[(1)(1) - (1)(0)] = \mathbf{-6}$$

$$\therefore \boldsymbol{det(A) = 2(12) - (-1)(-6) + 1(-6) - 0(6) = 12}$$

Since det(A) = 12 ≠ 0, the solution is unique.

Interpretation of Results

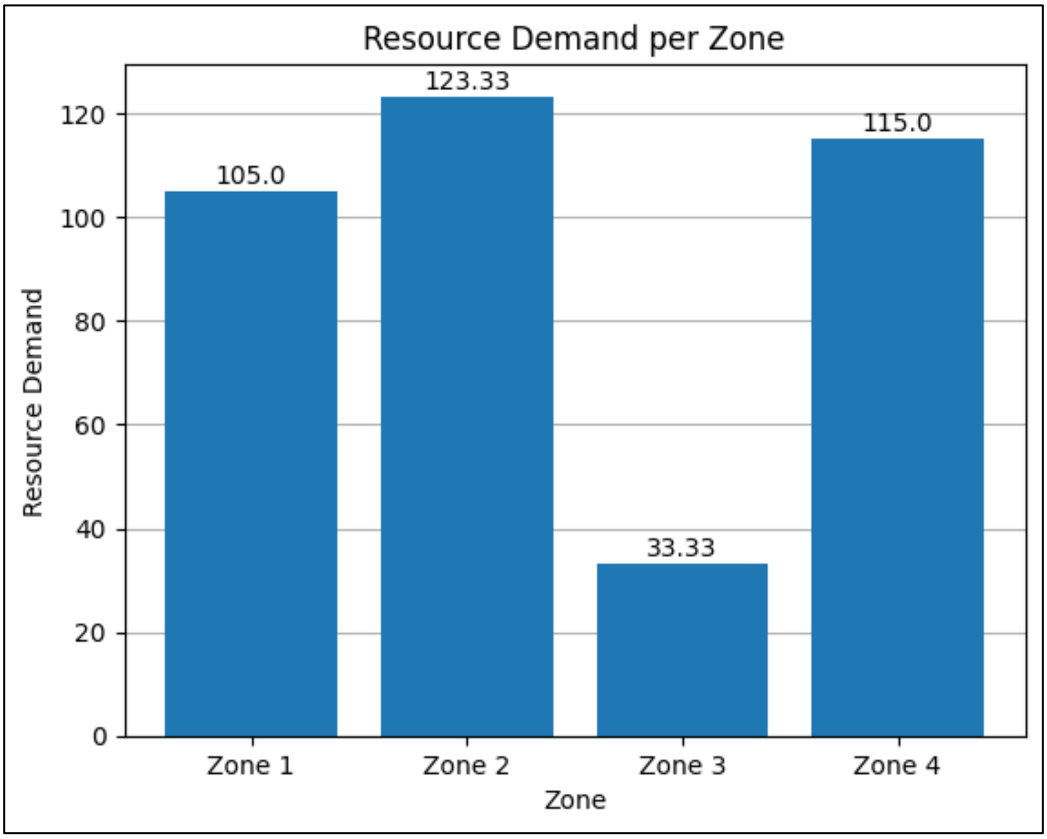

Figure 1: Demands across zones

The system has a unique solution with a stable equilibrium. Zone 2 has the highest demand at 123.33 units, followed by Zone 4 with 115 units and Zone 1 with 105 units, suggesting that these areas need more resources, especially Zone 2. Zone 3's demand is much lower at 33.33 units, indicating less pressure for resources.

Question 2: Calculus: Analysis of Demand Changes over Time

Introduction

This stage of the investigation examines how resource demand changes over time within a high-density residential area.

Approach

The demand is modelled using a cubic demand function:

$$D(t) = 4t^3 – 18t^2 + 24t + 90, \text{ where } t \geq 0 \text{ hours.}$$

Differentiation measures the instantaneous rate at which demand changes with respect to time. The first derivative identifies the periods during which demand increases or decreases, while the second derivative classifies each critical point (turning point) as a local maximum or minimum (Stewart, et al., 2020).

Critical points occur when:

$$D'(t) = 0,$$

and the second derivative test for finding the local maximum and minimum is as follows:

- **D''(t) > 0 → Local Minimum**
- **D''(t) < 0 → Local Maximum**

Solution

The differential of *D(t)* denoted as ,$\frac{dD}{dt}$ or **D'(t)** is defined as:

$$\frac{d}{dt}(t^n) = nt^{n-1}$$

such that:

$$D'(t) = \text{sum of the derivatives of each term}$$

This implies:

$$\frac{d}{dt}(4t^3) = 4 * \frac{d}{dt}(t^3) = 4(3t^{3-1}) = 12t^2 \quad \ldots \text{ first term}$$

$$\frac{d}{dt}(-18t^2) = -18 * \frac{d}{dt}(t^2) = -18(2t^{2-1}) = -36t \quad \ldots \text{ second term}$$

$$\frac{d}{dt}(24t) = 24 * \frac{d}{dt}(t) = 24(t^{1-1}) = 24 \quad \ldots \text{ third term}$$

$$\frac{d}{dt}(90) = 0 * \frac{d}{dt}(90t^0) = 0(90t^{0-1}) = 0 \quad \ldots \text{ fourth term}$$

$$\therefore \boldsymbol{D'(t) = 12t^2 - 36t + 24}$$

Factorizing the first derivative, *D'(t)*:

$$12(t^2 - 3t + 2) = 12(t - 1)\ (t - 2)$$

Solving for critical points, *D'(t)*:

$$12(t - 1)\ (t - 2) = 0$$

Using the Zero Product Property (Marecek, et al., 2025), the following equations are obtained:

$$12 = 0 \ldots (1)$$

$$t - 1 = 0 \ldots (2)$$

$$t - 2 = 0 \ldots (3)$$

Since 12 cannot equal 0, only equations (2) and (3) are considered, which result in ***t = 1*** and ***t = 2,*** respectively. The demand function *D(t)* has two critical points at **hours 1** and **2**.

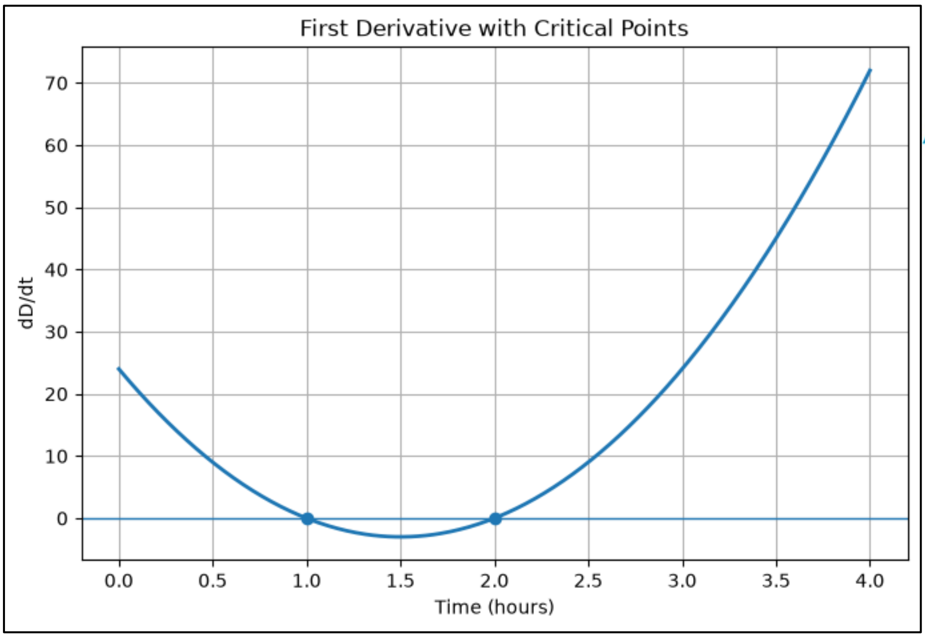

Figure 2: First derivative with critical points

Calculating D''(t):

$$\frac{d}{dt}(12t^2) = 12 * \frac{d}{dt}(t^2) = 12(2t^{2-1}) = 24t \text{ ... first term}$$

$$\frac{d}{dt}(-36t) = -36 * \frac{d}{dt}(t^1) = -36(1t^{1-1}) = -36 \text{ ... second term}$$

$$\frac{d}{dt}(24) = 0 * \frac{d}{dt}(24t^0) = 0(24t^{0-1}) = 0 \text{ ... third term}$$

$$\therefore \boldsymbol{D''(t) = 24t - 36}$$

Applying the second derivative test:

$$At\ \boldsymbol{t = 1}$$

$$\boldsymbol{D''(1)} = 24(1) - 36 = 24 - 36 = \underline{\boldsymbol{-12}}$$

Since $D''(1) < 0$, $\boldsymbol{t = 1}$ **is a local maximum**

$$At\ \boldsymbol{t = 2}$$

$$\boldsymbol{D''(}2) = 24(2) - 36 = 48 - 36 = \underline{\boldsymbol{12}}$$

Since $D''(2) > 0$, $\boldsymbol{t = 2}$ **is a local minimum**

To determine the demand at each critical point, the values of *t* are substituted into *D(t)*:

$$At\ \boldsymbol{t = 1}$$

$$\boldsymbol{D(1)} = 4(1)^3 - 18(1)^2 + 24(1) + 90 = 4 - 18 + 24 + 90 = \underline{\boldsymbol{100}}$$

$$At\ \boldsymbol{t = 2}$$

$$\boldsymbol{D(2)} = 4(2)^3 - 18(2)^2 + 24(2) + 90 = 4(8) - 18(4) + 24(2) + 90$$

$$= 32 - 72 + 48 + 90 = \underline{\boldsymbol{98}}$$

Checking the baseline demand level:

$$At\ \boldsymbol{t = 0}$$

$$D(0) = 4(0)^3 - 18(0)^2 + 24(0) + 90 = 0 - 0 + 0 + 90 = \underline{\boldsymbol{90}}$$

Sign Table

Interval	**Test Value**	**D'(t)**	**Sign**	**Interpretation**
$0 < t < 1$	0.5	$12(0.5)^2 - 36(0.5) + 24 = 9$	+	Demand increases
$1 < t < 2$	1.5	$12(1.5)^2 - 36(1.5) + 24 = -3$	-	Demand decreases
$t > 2$	3	$12(3)^2 - 36(3) + 24 = 24$	+	Demand increases

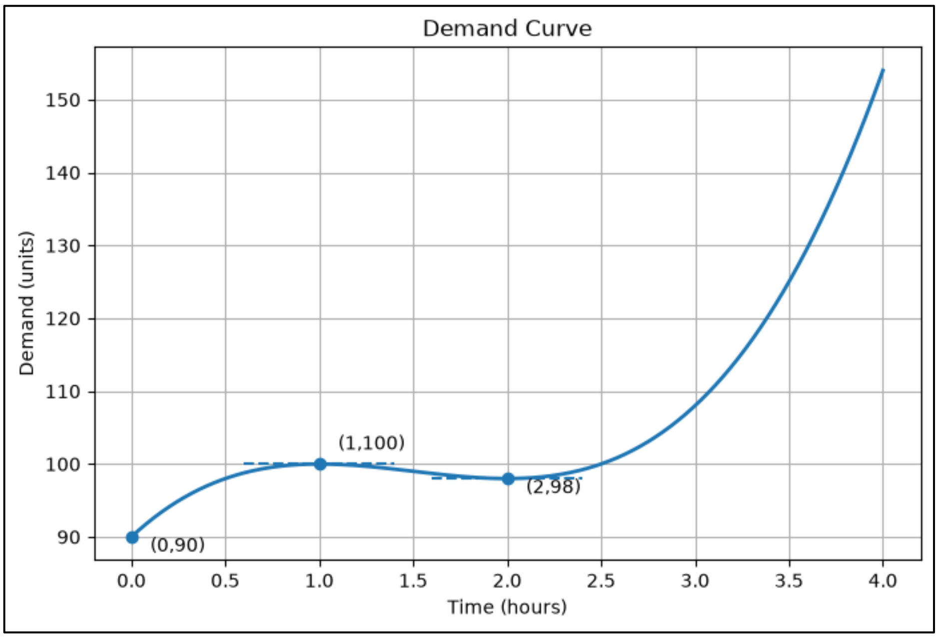

Figure 3: Demand over Time

Results

Result	Value	Interpretation
Initial demand	90 units	Starting demand level
Local maximum	(1, 100)	Highest demand; after one hour
Local minimum	(2, 98)	Lowest demand; before demand increases again
Increasing intervals	$0 < t < 1, t > 2$	Demand rises over time
Decreasing intervals	$1 < t < 2$	Demand falls temporarily

Interpretation of Results

The results above indicate that demand does not increase uniformly over time. Demand rises from 90 units to a local maximum of 100 units after one hour, before it declines slightly to 98 units after two hours. Beyond two hours, demand begins increasing again. Here, the local maximum represents the period of greatest pressure for infrastructure resources, while the local minimum indicates a temporary reduction in demand before a renewed increase.

Question 3: Probability: Analysis of Forecasting Errors

Introduction

This section examines the uncertainty of demand forecasts. Short-term errors, due to behavioural and environmental variability, are assumed to be normally distributed having a mean of 0 and a standard deviation of 5.

Approach

Because the forecasting error is normally distributed, the appropriate technique here is the **Z-score standardization** (Montgomery & Runger, 2020).

The general standardization formula is as follows**:**

$$\boldsymbol{Z = \frac{X - \mu}{\sigma}} \ldots (1)$$

where:

- X - forecasting error
- μ - mean of the distribution
- σ - standard deviation

After standardization, the probability of exceeding a threshold is read from Φ(z), which gives the probability that Z is at or above that threshold. A large deviation falls outside the empirical rule bands for normal distributions, i.e., errors exceeding 1, 2, or 3 standard deviations (Montgomery & Runger, 2020).

Solution

The forecasting error, E, is modelled as:

$$\boldsymbol{E \sim N(\mu = 0, \sigma = 5)}$$

Standardizing each threshold using the formula from equation (1) above:

- For X = 5:

$$Z = \frac{5 - 0}{5} = \mathbf{1}$$

- For X = 10:

$$Z = \frac{10 - 0}{5} = \mathbf{2}$$

- For X = 15:

$$Z = \frac{15 - 0}{5} = \mathbf{3}$$

According to the standard normal table:

- **Φ(1) = 0.8413**
- **Φ(2) = 0.9772**
- **Φ(3) = 0.9987**

The probability that the error overshoots the threshold on the positive side only can be calculated using:

P(E > x) = 1 − Φ(z)

Thus:

$$P(E > 5) = 1 - 0.8413 = 0.1587$$
$$P(E > 10) = 1 - 0.9772 = 0.0228$$
$$P(E > 15) = 1 - 0.9987 = 0.0013$$

Given that forecasting errors can be either an over-prediction (positive) or under-prediction (negative), whereby both are equally undesirable, the relevant probability is the two-tailed probability, computed as:

P(|E| > x) = 2 * [1 − Φ(z)] (Navidi, 2024)

$$P(|E| > 5) = 2(1 - 0.8413) = 0.3174$$
$$P(|E| > 10) = 2(1 - 0.9772) = 0.0456$$
$$P(|E| > 15) = 2(1 - 0.9987) = 0.0026$$

The results in their equivalent percentages and complementary within-threshold probability are shown below:

- For X = 5:

P(|E| > 5) = 0.3174 × 100 = 31.74%

$$P(|E| \leq 5) = 100\% - 31.74\% = 68.26\%$$

- For X = 10:

$$P(|E| > 10) = 0.0456 \times 100 = 4.56\%$$

$$P(|E| \leq 10) = 100\% - 4.56\% = 95.44\%$$

- For X = 15:

$$P(|E| > 15) = 0.0026 \times 100 = 0.26\%$$

$$P(|E| \leq 15) = 100\% - 0.26\% = 99.74\%$$

Results

Deviation Threshold	**Z-Score**	**$\Phi(z)$**	**P(E > x)**	**P(|E| > x)**	**P(|E| ≤ x)**
5 units (1σ)	1	0.8413	0.1587 (15.87%)	0.3174 (31.74%)	68.26%
10 units (2σ)	2	0.9772	0.0228 (2.28%)	0.0456 (4.56%)	95.44%
15 units (3σ)	3	0.9987	0.0013 (0.13%)	0.0026 (0.26%)	99.74%

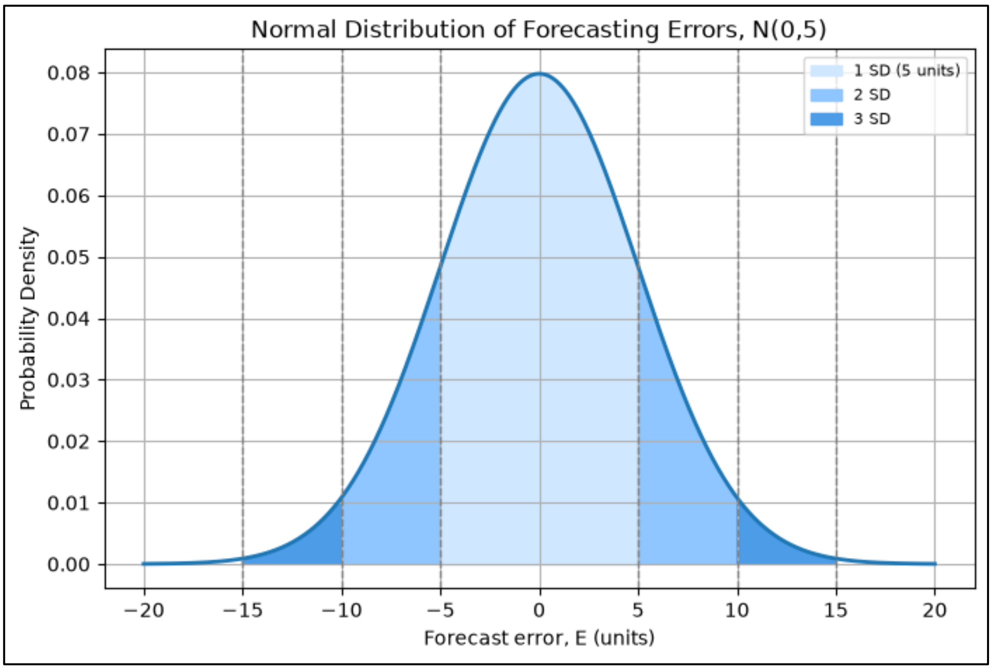

Figure 4: Distribution of forecasting errors

Interpretation of Results

Approximately 68% of daily forecasting errors are within ±5 units, 95% within ±10 units, and 99.7% within ±15 units (Illowsky & Dean, 2023). Errors that are larger than 10 units occur about 1 in every 22 days, and those over 15 units about 1 within 385 days. This supports risk management in that a 10-unit buffer covers for about 95% of errors, and 15 units for about 99.7%. However, this assumes that the errors follow a normal distribution and are independent from day to day, which, in reality, large deviations may cluster around events such as extreme weather.

Question 4: Statistics: Comparing Consumption Across Housing Types

Introduction

The final stage of this investigation checks whether daily resource use varies among the three residential building types.

Approach

Because three independent groups are being compared on a single continuous outcome of daily consumption, the appropriate technique to use here is a one-way Analysis of Variance (ANOVA) (Field, 2024).

The Between-Groups Sum of Squares and Shapiro-Wilk and Levene's tests will be conducted as well as an extra validation including well as Welch's ANOVA alongside classical ANOVA. Agreement between the two confirms reliability, and disagreement shall favour Welch's ANOVA (Navarro & Foxcroft, 2025).

Hypotheses:

H_0: μApartments = μTerraced = μDetached *(the population mean daily consumption is the same for all three housing types)*

H_1: at least one population mean differs from the others

Solution

Given the data:

- Apartments (n = 8): 16.8, 17.2, 16.5, 17.9, 18.1, 17.4, 16.9, 17.6
- Terraced houses (n = 10): 18.4, 18.9, 19.1, 18.7, 19.3, 18.6, 19.0, 18.8, 19.2, 18.5
- Detached houses (n = 6): 20.1, 19.7, 20.4, 19.9, 20.6, 20.2

The group mean is calculated as follows:

- $\bar{x}(Apartments) = \frac{16.8+17.2+16.5+17.9+18.1+17.4+16.9+17.6}{8} = \mathbf{17.30}$
- $\bar{x}(Terraced) = \frac{18.4+18.9+19.1+18.7+19.3+18.6+19.0+18.8+19.2+18.5}{10} = \mathbf{18.85}$

- $\bar{x}$*(Detached)* $= \frac{20.1+19.7+20.4+19.9+20.6+20.2}{6} =$ ***20.15***

The grand mean is calculated as follows:

$$\bar{x}\textit{(grand)} = \frac{138.4+188.5+120.9}{8+10+6} = \mathbf{18.6583}$$

(a) Normality: using the Shapiro-Wilk test

Housing Type	n	W statistic	p-value	Conclusion (α = 0.05)
Apartments	8	0.9733	0.9222	p > 0.05 → normality holds
Terraced	10	0.9702	0.8924	p > 0.05 → normality holds
Detached	6	0.9902	0.9897	p > 0.05 → normality holds

In every group, p > 0.05, so there is no evidence against normality within any of the three housing types.

(b) Homogeneity of variance using Levene's test, applied across all three groups simultaneously:

$$\textit{Levene statistic} = \mathbf{2.4889}, \quad p = \mathbf{0.1071}$$

Since p (0.1071) > 0.05, there is no evidence of significant variance differences across groups, supporting the use of standard (Fisher) ANOVA. Welch's ANOVA shall also be calculated as a robustness check due to the unequal sample sizes in the housing types.

Calculating the Between-Groups Sum of Squares (SSB):

SSB measures each group's mean distance from the grand mean, weighted by sample size n_i, accounting for the terraced-house group's (n=10) greater contribution relative to the detached-house group (n=6).

$$\boldsymbol{SSB = \sum n_i(\bar{x}_i - \bar{x}grand)^2}$$

$$SSB = 8(17.30-18.6583)^2 + 10(18.85-18.6583)^2 + 6(20.15-18.6583)^2$$

$$SSB = 8(-1.3583)^2 + 10(0.1917)^2 + 6(1.4917)^2$$

$$SSB = 8(1.8451) + 10(0.0367) + 6(2.2251)$$

$$SSB = 14.7605 + 0.3674 + 13.3505 = \mathbf{28.4784}$$

Calculating the Within-Groups Sum of Squares (SSW):

SSW measures how far each individual observation is from its own group's mean, which requires computing every squared deviation:

Apartments (mean = 17.30):

x_i	x_i − mean	(x_i − mean)²
16.8	−0.50	0.2500
17.2	−0.10	0.0100
16.5	−0.80	0.6400
17.9	0.60	0.3600
18.1	0.80	0.6400
17.4	0.10	0.0100
16.9	−0.40	0.1600
17.6	0.30	0.0900
Sum	0.00	2.1600

Terraced houses (mean = 18.85):

x_i	x_i − mean	(x_i − mean)²
18.4	−0.45	0.2025
18.9	0.05	0.0025
19.1	0.25	0.0625
18.7	−0.15	0.0225
19.3	0.45	0.2025
18.6	−0.25	0.0625
19.0	0.15	0.0225
18.8	−0.05	0.0025
19.2	0.35	0.1225
18.5	−0.35	0.1225
Sum	0.00	0.8250

Detached houses (mean = 20.15):

x_i	x_i − mean	(x_i − mean)²
20.1	−0.05	0.0025
19.7	−0.45	0.2025
20.4	0.25	0.0625
19.9	−0.25	0.0625
20.6	0.45	0.2025
20.2	0.05	0.0025
Sum	0.00	0.5350

Summing the three group totals:

$$SSW = 2.1600 + 0.8250 + 0.5350 = \mathbf{3.5200}$$

Calculating the Total Sum of Squares (SST) as a check:

$$SST = SSB + SSW = 28.4784 + 3.5200 = \mathbf{31.9984}$$

Calculating degrees of freedom with k = 3 groups and N = 24 total observations:

$$\boldsymbol{df(between)} = k - 1 = 3 - 1 = \mathbf{2}$$

$$\boldsymbol{df(within)} = N - k = 24 - 3 = \mathbf{21}$$

$$\boldsymbol{df(total)} = N - 1 = 24 - 1 = \mathbf{23}$$

Calculating the mean squares using:

Mean square = sum of squares ÷ its degrees of freedom:

$$MSB = SSB / df(between) = 28.4784 / 2 = \mathbf{14.2392}$$

$$MSW = SSW / df(within) = 3.5200 / 21 = \mathbf{0.1676}$$

Calculating the F-statistic:

$$F = MSB / MSW = 14.2392 / 0.1676 = \mathbf{84.95}$$

Comparing against the critical F-value, at a significance level of $\alpha = 0.05$, with $df1 = 2$ and $df2 = 21$, the critical value from the F-distribution table is:

$$F_{critical}(2,21) = \mathbf{3.4668}$$

Since the calculated F-statistic (84.95) is greater than the critical value (3.47):

$$F_{calculated} = 84.95 \; > \; F_{critical} = 3.47$$

H_0 is rejected as there is strong statistical evidence that the mean daily consumption differs across the three housing types.

To express how much of the total variation in consumption is by housing type, eta-squared is calculated:

$$\eta^2 = SSB / SST = 28.4784 / 31.9984 = \mathbf{0.8900}$$

This means approximately 89% of the variation in daily consumption across the sample is explained by knowing which housing type a dwelling belongs to (Pallant, 2020).

Welch's ANOVA is now calculated; it weights each group by:

$$\boldsymbol{w_i = n_i / s_i^2}$$

giving w(Apartments) = 25.66, w(Terraced) = 109.09, w(Detached) = 56.05 (using each group's sample variance). These weights are combined into a weighted grand mean, and the resulting statistic is:

$$\boldsymbol{F(Welch) = 71.66, \quad df1 = 2, \quad df2 = 11.75}$$

$$\boldsymbol{F_{critical,\ Welch}(2, 11.75) = 3.91}$$

Since F(Welch) (71.66) > Fcritical (3.91), Welch's ANOVA reaches the same conclusion as the classical ANOVA: H_0 is rejected.

Because the normality assumption holds in every group, Levene's test shows no significant difference in variances, and both the classical ANOVA and Welch's ANOVA reject H_0 with very similar strength, the conclusion that mean consumption differs by housing type is reliable and not an artifact of the unequal sample sizes.

Results

Table 1: Assumption checks

Test	Purpose	Statistic	p-value	Conclusion
Shapiro–Wilk (Apartments)	Normality	W = 0.9733	0.9222	Normal
Shapiro–Wilk (Terraced)	Normality	W = 0.9702	0.8924	Normal
Shapiro–Wilk (Detached)	Normality	W = 0.9902	0.9897	Normal
Levene's test	Homogeneity of variance	2.4889	0.1071	Variances equal

Table 2: ANOVA summary

Source of Variation	SS	df	MS	F	F-crit (α=0.05)	Decision
Between groups	28.4784	2	14.2392	84.95	3.47	Reject H_0
Within groups	3.5200	21	0.1676	-	-	-
Total	31.9984	23	-	-	-	-

Table 3: Robustness check for unequal sample sizes

Test	F-statistic	df1	df2	F-critical	Decision
Classical (Fisher) ANOVA	84.95	2	21	3.47	Reject H_0
Welch's ANOVA	71.66	2	11.75	3.91	Reject H_0

Table 4: Descriptive statistics

Housing Type	n	Mean (units)	Standard Deviation
Apartments	8	17.30	0.56
Terraced houses	10	18.85	0.30
Detached houses	6	20.15	0.33

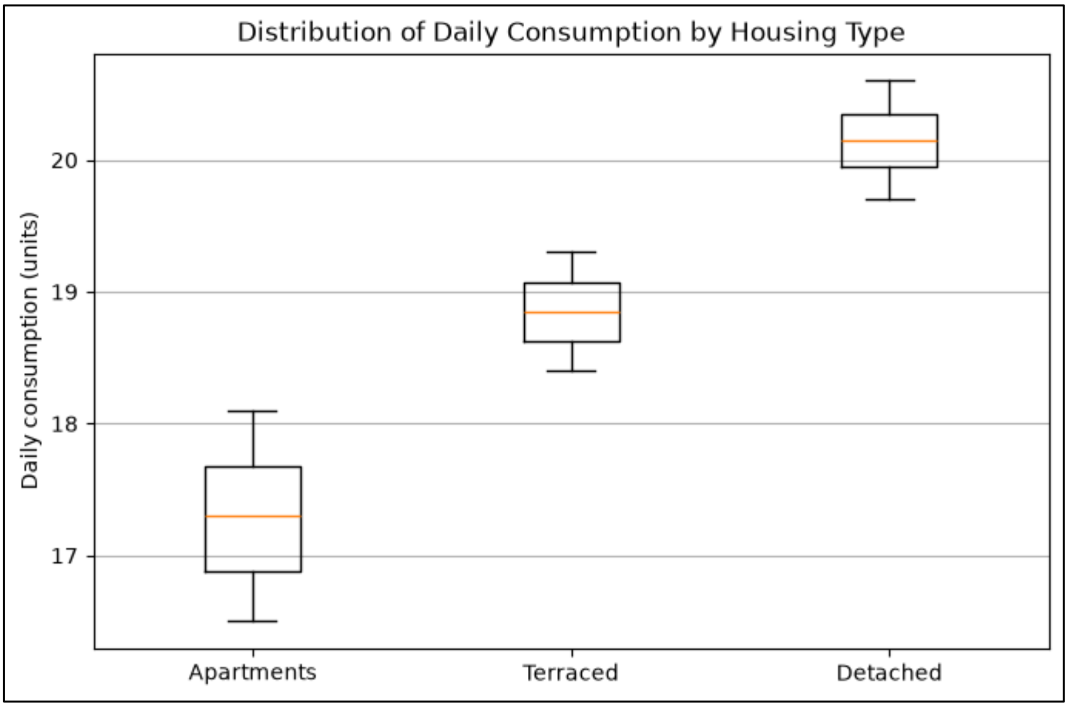

Figure 5: Distribution of Daily Consumption by Housing Type

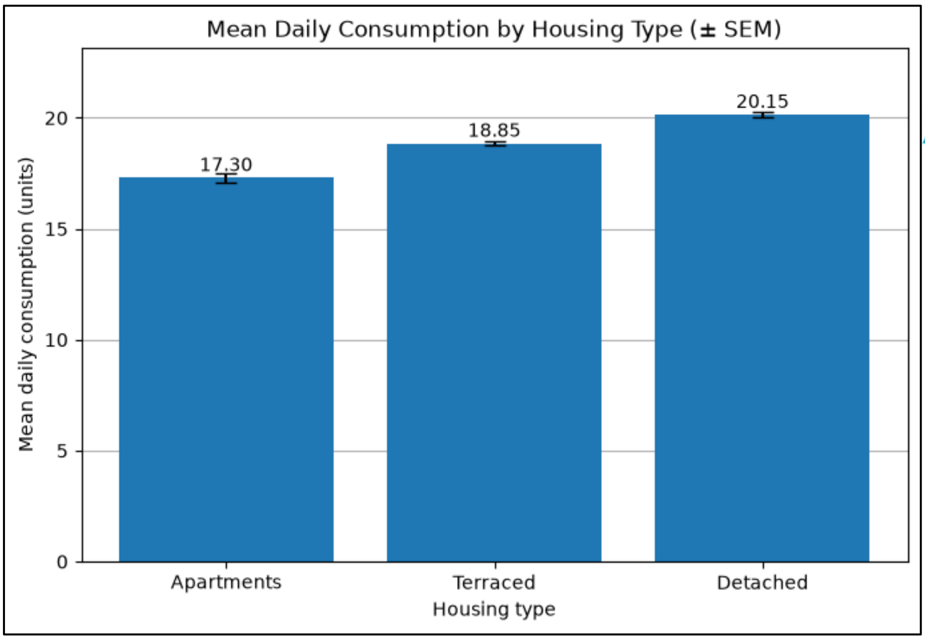

Figure 6: Mean Daily Consumption by Housing Type (± Standard Error of the Mean)

Interpretation of Results

The one-way ANOVA indicates significant differences among the three housing types (Apartments = 17.30, Terraced = 18.85, Detached = 20.15 units), with a high F-value (84.95 vs. critical value of 3.47). Consumption increases with dwelling size, explaining about 89% of the variation in the data. Despite the unequal sample sizes, normality and similar variances were confirmed and supported by Welch's ANOVA, which allowed for high confidence in the conclusions. This means that targeted interventions per housing type are recommended. Two limitations are, however, noted: the ANOVA confirms differences but not all pairwise ones, needing further testing; the small detached-house sample (n=6) suggests larger samples should be provided for future studies.

Critical Evaluation

Assumptions and Limitations of the Models

Each of the four analyses relies on assumptions that simplify a real city, but these are however still simplifications. The linear algebra model assumes fixed relationships between the four zones, meaning one zone's influence on another never changes. In reality, these relationships change over time as factors like population, buildings, and infrastructure evolve. The calculus model assumes that the demand follows a smooth curve, which describes short-term changes but suggests indefinite growth if extended indefinitely, which is unrealistic for long-term forecasts. The probability model assumes forecasting errors are uniformly distributed, but events such as weather, holidays, or behavioral changes can cause larger errors, so its figures are best seen as averages and not guarantees. The housing type comparison, on the other hand, assumes consistent consumption within each type and similar values from the calculation results across apartments, terraced, and detached houses, verified directly but with less robustness for detached houses due to a smaller sample.

Reliability of Conclusions Under Uncertainty

Some conclusions are as certain as the mathematics, like the demand levels for four zones following directly from given equations. Others are about likelihood, indicating how often these types of errors are expected over time. The housing comparison was carefully handled due to differences in sample size with the use of two tests for validation, and both of them agreed, supporting the conclusion that consumption differs by housing type.

Practical Implications for Urban Policy and Planning

The zone-level equilibrium shows that Zone 2 has the highest steady demand among the four zones, at roughly 123 units, compared with Zone 3's much lower 33 units; so if resources must be prioritized, Zone 2 should come first.

The time-based analysis shows that demand within a high-density area peaks around the one-hour mark (rising to 100 units from a starting point of 90) before easing slightly by the two-hour mark. This therefore indicates to operational teams when during the day extra resources are needed.

The forecasting-error analysis showed that holding a buffer of about 10 units would be enough to cover for demand on roughly 95 out of every 100 days, and a buffer of about 15 units would cover demand on roughly 99.7 out of every 100 days. This gives the local authorities a concrete way to measure the cost of holding spare capacity against the risk of running short.

Finally, the comparison across housing types shows that detached houses consume more on average (around 20 units a day) than terraced houses (around 19 units) and apartments (around 17 units), which supports prioritizing detached houses for efficiency measures.

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Appendix

```
import numpy as np
import matplotlib.pyplot as plt
import sympy as sp
import scipy.stats as stats

plt.style.use('default')

# 1. calculation
A = np.array([
    [2, -1, 1, 0],
    [-1, 3, 0, -1],
    [1, 1, 2, -1],
    [0, -1, 1, 2]
], dtype=float)

b = np.array([120, 150, 180, 140], dtype=float)

x = np.linalg.solve(A, b)
result = [round(i, 2) for i in x]
print(f"""Result:
  x1 = {result[0]}
  x2 = {result[1]}
  x3 = {result[2]}
  x4 = {result[3]}""")

# 2. uniqueness check
det = np.linalg.det(A)
print(f'Det(A) = {det}')

# 3. visualization
x = np.array(["Zone 1", "Zone 2", "Zone 3", "Zone 4"])
y = result
plt.bar(x, y,)
for i in range(len(x)):
    plt.text(x[i], y[i] + 1.5, str(y[i]), ha='center')
plt.xlabel("Zone")
plt.ylabel("Resource Demand")
plt.title("Resource Demand per Zone")
plt.grid(axis='y')
plt.gca().set_axisbelow(True)
plt.show()



t = sp.symbols('t')
d = 4*t**3 - 18*t**2 + 24*t + 90
```

```
print('Demand function:')
sp.pprint(d)

d1 = sp.diff(d, t)
print('\nFirst derivative:')
sp.pprint(d1)

print('\nFactorized first derivative:')
sp.pprint(sp.factor(d1))

critical_points = sp.solve(d1, t)
print('\nCritical points:', critical_points)

d2 = sp.diff(d1, t)
print('\nSecond derivative:')
sp.pprint(d2)

print('\nSecond derivative test:')
for point in critical_points:
    print(f"D"({point}) =", d2.subs(t, point))

print("\nDemand at Critical Points:")
for point in critical_points:
    print(f"D({point}) =", d.subs(t, point))

print("\nBaseline Demand:\n")
print("D(0) =", d.subs(t, 0))

for item in [d, d1, d2]:
    print(item)

# Time values
t = np.linspace(0, 4, 400)

# Functions
D = 4*t**3 - 18*t**2 + 24*t + 90
D1 = 12*t**2 - 36*t + 24
D2 = 24*t - 36

# Critical points
critical_t = np.array([0, 1, 2])
critical_D = 4*critical_t**3 - 18*critical_t**2 + 24*critical_t + 90

# demand function
plt.figure(figsize=(8,5))
plt.plot(t, D, linewidth=2)
plt.scatter(critical_t, critical_D, zorder=3)
plt.hlines(100, 0.6, 1.4, linestyles='dashed')
```

```
plt.hlines(98, 1.6, 2.4, linestyles='dashed')
plt.annotate("(1,100)", (1,100), xytext=(1.1,102))
plt.annotate("(2,98)", (2,98), xytext=(2.1,96))
plt.annotate("(0,90)", (0,90), xytext=(0.1,88))
plt.xlabel("Time (hours)")
plt.ylabel("Demand (units)")
plt.title("Demand Curve")
plt.grid(True)
plt.show()

# first derivative with critical points
plt.figure(figsize=(8,5))
plt.plot(t, D1, linewidth=2)
plt.axhline(0, linewidth=1)
plt.scatter([1,2],[0,0], zorder=3)
plt.xlabel("Time (hours)")
plt.ylabel("dD/dt")
plt.title("First Derivative with Critical Points")
plt.grid(True)
plt.show()



mu, sigma = 0, 5
thresholds = [5, 10, 15]

print(f"{'x (units)':>10} | {'z':>4} | {'Phi(z)':>8} | {'P(E>x)':>8} | {'P(|E|>x)':>10}")
for x in thresholds:
    z = x / sigma
    phi = stats.norm.cdf(z)
    one_tail = 1 - phi
    two_tail = 2 * one_tail
    print(f"{x:>10} | {z:>4.1f} | {phi:>8.4f} | {one_tail:>8.4f} | {two_tail:>10.4f}")



x = np.linspace(-20, 20, 1000)
pdf = stats.norm.pdf(x, mu, sigma)

plt.figure(figsize=(8,5))
plt.plot(x, pdf, linewidth=2, color='tab:blue')

plt.fill_between(x, pdf, where=(x>=-5)&(x<=5), color='#cfe8ff', label='1 SD (5 units)')
plt.fill_between(x, pdf, where=((x>5)&(x<=10))|((x<-5)&(x>=-10)), color='#8fc6ff', label='2 SD')
plt.fill_between(x, pdf, where=((x>10)&(x<=15))|((x<-10)&(x>=-15)),
color='#4c9ce8', label='3 SD')

for k in [5,10,15]:
    plt.axvline(k, linestyle='dashed', color='grey', linewidth=1)
```

```
    plt.axvline(-k, linestyle='dashed', color='grey', linewidth=1)

plt.xlabel("Forecast error, E (units)")
plt.ylabel("Probability Density")
plt.title("Normal Distribution of Forecasting Errors, N(0,5)")
plt.legend(fontsize=8, loc='upper right')
plt.grid(True)
plt.show()



# 1. Data
apartments = np.array([16.8, 17.2, 16.5, 17.9, 18.1, 17.4, 16.9, 17.6])
terraced   = np.array([18.4, 18.9, 19.1, 18.7, 19.3, 18.6, 19.0, 18.8, 19.2, 18.5])
detached   = np.array([20.1, 19.7, 20.4, 19.9, 20.6, 20.2])

groups = [apartments, terraced, detached]
labels = ["Apartments", "Terraced", "Detached"]

# 2. Descriptive statistics
means = [g.mean() for g in groups]
ns = [len(g) for g in groups]
grand_mean = np.concatenate(groups).mean()

for lab, g, m in zip(labels, groups, means):
    print(f"{lab}: n={len(g)}, mean={m:.4f}, sd={g.std(ddof=1):.4f}")
print(f"\nGrand mean = {grand_mean:.4f}")



# 3. One-way ANOVA - verification of manual calculation
F_stat, p_value = stats.f_oneway(apartments, terraced, detached)
print(f"F statistic = {F_stat:.4f}")
print(f"p-value     = {p_value:.6e}")

# 4. Manual breakdown: SSB, SSW, df, MSB, MSW, F
ssb = sum(len(g) * (m - grand_mean)**2 for g, m in zip(groups, means))
ssw = sum(((g - m)**2).sum() for g, m in zip(groups, means))
sst = ssb + ssw

df_between = len(groups) - 1
df_within = len(np.concatenate(groups)) - len(groups)

msb = ssb / df_between
msw = ssw / df_within
F_manual = msb / msw

f_crit = stats.f.ppf(0.95, df_between, df_within)
eta_sq = ssb / sst
```

```
print(f"\nSSB={ssb:.4f}, SSW={ssw:.4f}, SST={sst:.4f}")
print(f"df_between={df_between}, df_within={df_within}")
print(f"MSB={msb:.4f}, MSW={msw:.4f}")
print(f"F (manual) = {F_manual:.4f}")
print(f"F critical (alpha=0.05) = {f_crit:.4f}")
print(f"eta^2 (effect size) = {eta_sq:.4f}")

# Assumption checks
print("Shapiro-Wilk normality test (per group):")
for lab, g in zip(labels, groups):
    stat, p = stats.shapiro(g)
    print(f"  {lab}: W={stat:.4f}, p={p:.4f}")

print("\nLevene's test for homogeneity of variance (all 3 groups):")
stat, p = stats.levene(*groups)
print(f"  Levene statistic={stat:.4f}, p={p:.4f}")

# Welch's ANOVA - robust to unequal variances and unequal sample sizes
ns = np.array([len(g) for g in groups])
grp_means = np.array([g.mean() for g in groups])
grp_vars = np.array([g.var(ddof=1) for g in groups])
k = len(groups)

w = ns / grp_vars
sum_w = w.sum()
weighted_grand_mean = (w*grp_means).sum() / sum_w

numerator = (w*(grp_means-weighted_grand_mean)**2).sum() / (k-1)
term = ((1-w/sum_w)**2/(ns-1)).sum()
denominator = 1 + (2*(k-2)/(k**2-1))*term

F_welch = numerator/denominator
df1_welch = k-1
df2_welch = (k**2-1)/(3*term)
p_welch = 1 - stats.f.cdf(F_welch, df1_welch, df2_welch)
f_crit_welch = stats.f.ppf(0.95, df1_welch, df2_welch)

print(f"\nWelch's ANOVA: F={F_welch:.4f}, df1={df1_welch}, df2={df2_welch:.4f}")
print(f"F critical (Welch)={f_crit_welch:.4f}, p={p_welch:.6e}")


# 5. Visualisation - boxplot of consumption distribution by housing type
plt.figure(figsize=(8,5))
plt.boxplot([apartments, terraced, detached], tick_labels=labels)
plt.ylabel("Daily consumption (units)")
plt.title("Distribution of Daily Consumption by Housing Type")
plt.grid(axis='y')
```

```
plt.gca().set_axisbelow(True)
plt.show()

# 6. Visualisation - bar chart of group means with standard error bars
sems = [g.std(ddof=1) / np.sqrt(len(g)) for g in groups]

plt.figure(figsize=(8,5))
plt.bar(labels, means, yerr=sems, capsize=6, color='tab:blue')
for i in range(len(labels)):
    plt.text(i, means[i] + 0.3, f"{means[i]:.2f}", ha='center')
plt.xlabel("Housing type")
plt.ylabel("Mean daily consumption (units)")
plt.title("Mean Daily Consumption by Housing Type (\u00b1 SEM)")
plt.grid(axis='y')
plt.gca().set_axisbelow(True)
plt.ylim(0, max(means) + 3)
plt.show()
```